ZOHAIR ALAM

Full-Stack Software Engineer

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SUMMARY

Full-Stack Software Engineer with hands-on experience building web applications using React.js and Java Spring Boot, plus applied project experience in NLP and Computer Vision. Comfortable designing RESTful APIs, working with relational databases, and containerizing services with Docker. Strong conceptual grounding in machine learning fundamentals (transformer architectures, model evaluation, coreference resolution, object detection) from university coursework and applied course projects; still growing hands-on depth on the ML engineering side.

EXPERIENCE

Front-End Engineer Intern — *Sunshine Digital Solutions, Karachi, Pakistan* May 2026 – Jul 2026

- Built and shipped multiple client-facing websites using React.js and Tailwind CSS
- Translated Figma designs into pixel-perfect, responsive front-end implementations
- Applied web performance optimization techniques to improve load times and rendering efficiency

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), Python, SQL, HTML5, CSS3

Backend & Frameworks: Spring Boot, Spring MVC, Spring Data JPA/Hibernate, RESTful APIs, Microservices, Maven

Frontend: React.js, Context API, JavaScript, Tailwind CSS, Material UI

AI / ML / NLP (applied coursework): Tokenization, Embeddings, Attention Mechanisms, Transformer Architectures (BERT/RoBERTa), Model Fine-Tuning & Evaluation (Precision/Recall/F1), Hugging Face, PyTorch basics, OpenCV, Object Detection (YOLO)

Databases: MySQL, PostgreSQL, MongoDB, Database Indexing & Normalization

Tooling: Docker, Git/GitHub, Postman

FEATURED PROJECTS

AI-Assisted Finance Tracking Platform — *React.js, Spring Boot, Python/FastAPI, MySQL, Docker, REST APIs*

- Built a full-stack finance tracker with a React frontend and a Spring Boot backend handling core transactions, auth, and data persistence via Spring Data JPA and MySQL
- Added a Python/FastAPI microservice that parses bank-statement text with an NLP classification model to auto-categorize transactions and power spending-insight dashboards
- Containerized the frontend, backend, and database with Docker Compose for a consistent local/staging setup

NLP Coreference Resolution (Qalb Dataset) — *Python, PyTorch, Hugging Face Transformers, Scikit-learn*

- Fine-tuned a transformer-based language model on the Qalb dataset to detect and resolve ambiguous pronoun-reference chains in text, working from Hugging Face base checkpoints
- Built preprocessing and tokenization pipelines to convert raw text into sequence-labeling tensors with attention masks
- Compared baseline vs. fine-tuned checkpoints using Precision, Recall, and F1, and reviewed misclassified cases to identify weak spots in long-distance pronoun references

Real-Time Classroom Object Detection — *Python, OpenCV, YOLO/PyTorch*

- Built a computer-vision pipeline to detect and classify classroom objects (desks, chairs, whiteboards, devices) from video and static images
- Prepared a custom training set with bounding-box labeling and augmentation (rotation, brightness, noise) to improve robustness across lighting conditions
- Ran inference with a pre-trained YOLO backbone via OpenCV and evaluated results using mean Average Precision (mAP) and inference frame rate

Campus Analytics & Operations Dashboard — *Python, Streamlit, Pandas, Plotly*

- Built an interactive dashboard to clean, explore, and visualize university operational and demographic data
- Implemented temporal filtering, categorical grouping, and interactive visuals (heatmaps, time-series charts) to surface trends for reporting

ADDITIONAL PROJECTS

- XenonCreatives & VR Startup Websites — built and deployed marketing sites using React.js, Tailwind CSS, and Firebase
- Android Apps (pet-supplies e-commerce, shoe store, to-do list) — native apps in Kotlin/Java with XML UIs, Firebase, SQLite, and Room Database

EDUCATION

B.E. Software Engineering — *Iqra University, Main Campus, Karachi, Pakistan* Mar 2022 – Jul 2026

GPA: 3.52 | Relevant coursework: Machine Learning, Natural Language Processing, Data Science, Cybersecurity & Data Protection